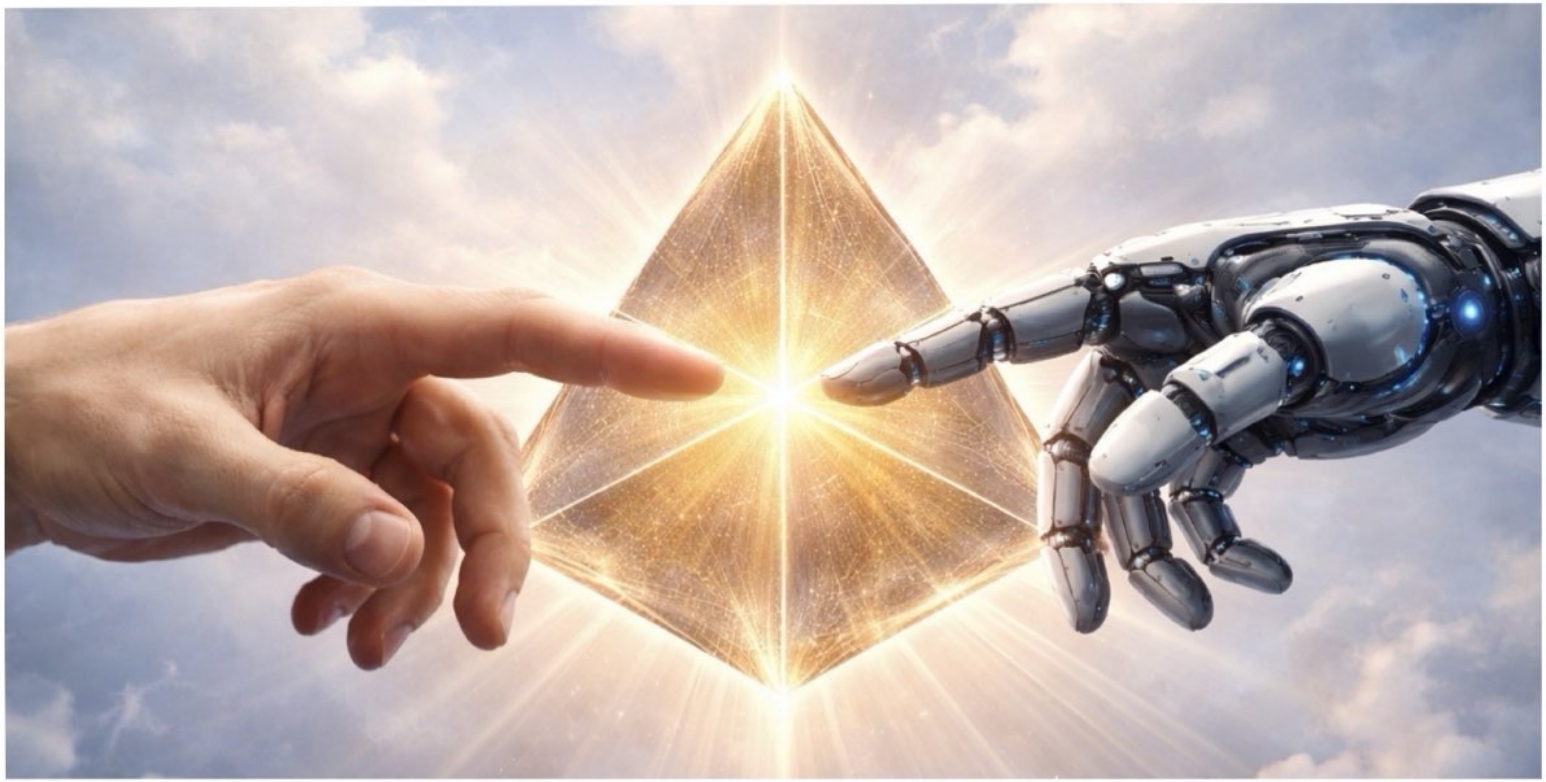
**StegVerse Governed Execution Runtime**
**Deterministic Governed Workflow Demonstration**
**Execution Transcript — v1.1**



Command: ./stegverse demo

Environment: Windows 11 • Git Bash

Repository:
https://github.com/stegverse-org/stegverse-demo-suite

Reproducible runtime demonstration
Governed execution + receipt-chain verification

## Execution Overview

This document contains a reproducible execution transcript of the StegVerse governed runtime demonstration.

The runtime enforces admissible state transitions where each step produces an execution receipt that advances system state and unlocks the next governed artifact.

The transcript demonstrates the following workflow:

- repository clone
- runtime inspection
- governed execution
- receipt-chain generation
- final artifact unlock validation

## Result

The runtime enforces admissible state transitions where each step produces an execution receipt that unlocks the next governed artifact.

This transcript demonstrates runtime enforcement of admissibility, state transitions, and artifact access within the execution environment.

# Appendix A — Execution Transcript

```
rigel@Chopper-00 MINGW64 ~
$ cd /c/Users/rigel/Documents

rigel@Chopper-00 MINGW64 ~/Documents
$ ls
'My Music'@  'My Pictures'@  'My Videos'@

rigel@Chopper-00 MINGW64 ~/Documents
$ git clone https://github.com/StegVerse/stegverse-demo-suite.git
Cloning into 'stegverse-demo-suite'...
info: please complete authentication in your browser...
remote: Repository not found.
fatal: repository 'https://github.com/StegVerse/stegverse-demo-suite.git/' not found

rigel@Chopper-00 MINGW64 ~/Documents
$ git clone https://github.com/StegVerse-org/stegverse-demo-suite.git
Cloning into 'stegverse-demo-suite'...
remote: Enumerating objects: 267, done.
remote: Counting objects: 100% (70/70), done.
remote: Compressing objects: 100% (64/64), done.
remote: Total 267 (delta 39), reused 4 (delta 4), pack-reused 197 (from 1)
Receiving objects: 100% (267/267), 105.83 KiB | 967.00 KiB/s, done.
Resolving deltas: 100% (109/109), done.

rigel@Chopper-00 MINGW64 ~/Documents
$ ls
'My Music'@  'My Pictures'@  'My Videos'@   stegverse-demo-suite/

rigel@Chopper-00 MINGW64 ~/Documents
$ cd stegverse-demo-suite

rigel@Chopper-00 MINGW64 ~/Documents/stegverse-demo-suite (main)
$ ls
docs/  engine/  README.md  RUNTIME_WRAPPER_NOTES.txt  SMOKE_TEST_RUNTIME_WRAPPER.json  stegverse*  workflow/

rigel@Chopper-00 MINGW64 ~/Documents/stegverse-demo-suite (main)
$ ls docs
architecture.md  doc2_demo2.md  doc4_demo4.md           RUN_DEMO_INSTRUCTIONS.md
doc1_demo1.md    doc3_demo3.md  doc5_system_summary.md

rigel@Chopper-00 MINGW64 ~/Documents/stegverse-demo-suite (main)
```

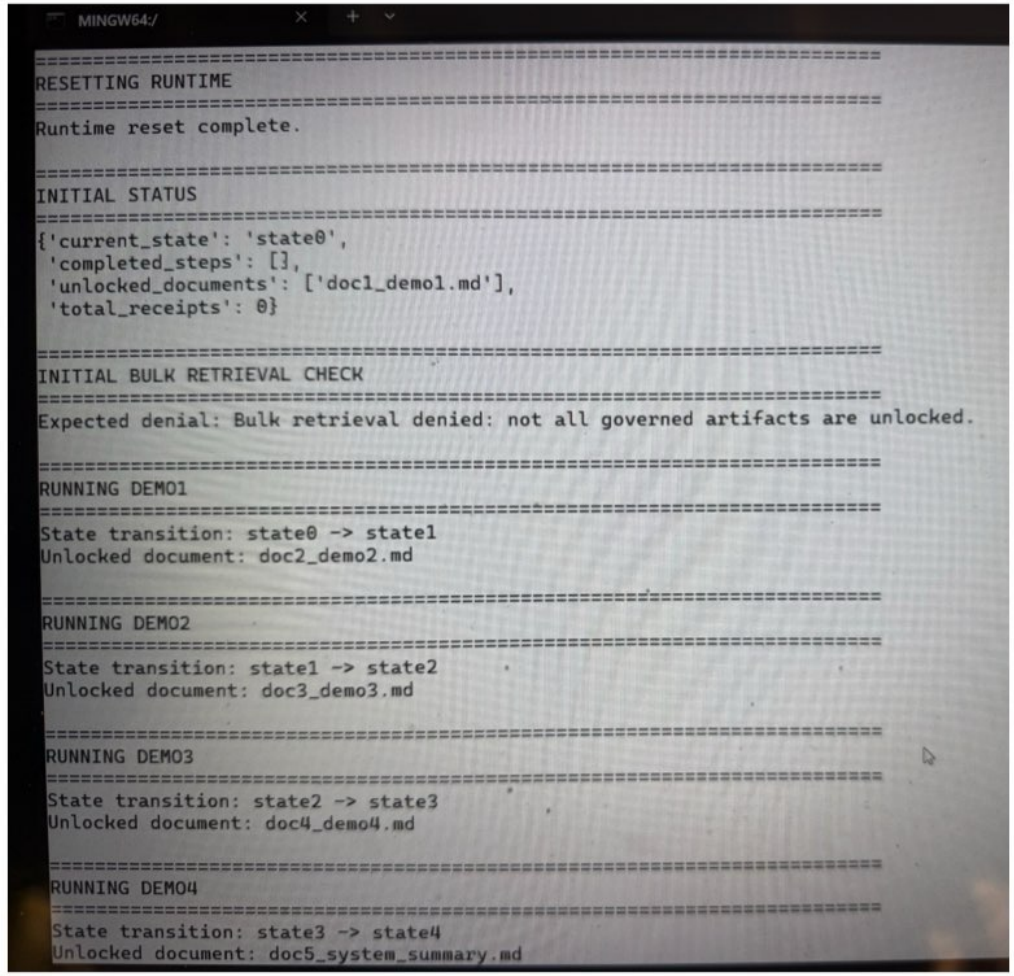
1) Repository Clone

# Appendix A — Execution Transcript

```
MINGW64:/

rigel@Chopper-00 MINGW64 ~/Documents/stegverse-demo-suite (main)
$ ls engine
__pycache__/  doc_gate.py  run_demo.py  stegverse_cli.py

rigel@Chopper-00 MINGW64 ~/Documents/stegverse-demo-suite (main)
$ ls workflow
manifest.json

rigel@Chopper-00 MINGW64 ~/Documents/stegverse-demo-suite (main)
$ ./stegverse help
StegVerse Runtime
----------------------------------------
Commands: help, explain, status, list, tree, run <step_id>, get <document>, receipts, bulk, reset, demo
Examples:
  stegverse.cmd demo   (Windows)
  ./stegverse demo     (Unix-like shells)

rigel@Chopper-00 MINGW64 ~/Documents/stegverse-demo-suite (main)
$ ./stegverse explain

StegVerse Runtime
----------------------------------------
current state: state0
completed steps: none
receipts: 0

next admissible steps:
  - demo1

unlocked artifacts:
  - doc1_demo1.md

locked artifacts:
  - doc2_demo2.md
  - doc3_demo3.md
  - doc4_demo4.md
  - doc5_system_summary.md

rigel@Chopper-00 MINGW64 ~/Documents/stegverse-demo-suite (main)
$ ./stegverse demo
```

2) Runtime Inspection

# Appendix A — Execution Transcript



```
MINGW64:/
================================================================================
RESETTING RUNTIME
================================================================================
Runtime reset complete.

================================================================================
INITIAL STATUS
================================================================================
{'current_state': 'state0',
 'completed_steps': [],
 'unlocked_documents': ['doc1_demo1.md'],
 'total_receipts': 0}

================================================================================
INITIAL BULK RETRIEVAL CHECK
================================================================================
Expected denial: Bulk retrieval denied: not all governed artifacts are unlocked.

================================================================================
RUNNING DEMO1
================================================================================
State transition: state0 -> state1
Unlocked document: doc2_demo2.md

================================================================================
RUNNING DEMO2
================================================================================
State transition: state1 -> state2
Unlocked document: doc3_demo3.md

================================================================================
RUNNING DEMO3
================================================================================
State transition: state2 -> state3
Unlocked document: doc4_demo4.md

================================================================================
RUNNING DEMO4
================================================================================
State transition: state3 -> state4
Unlocked document: doc5_system_summary.md
```

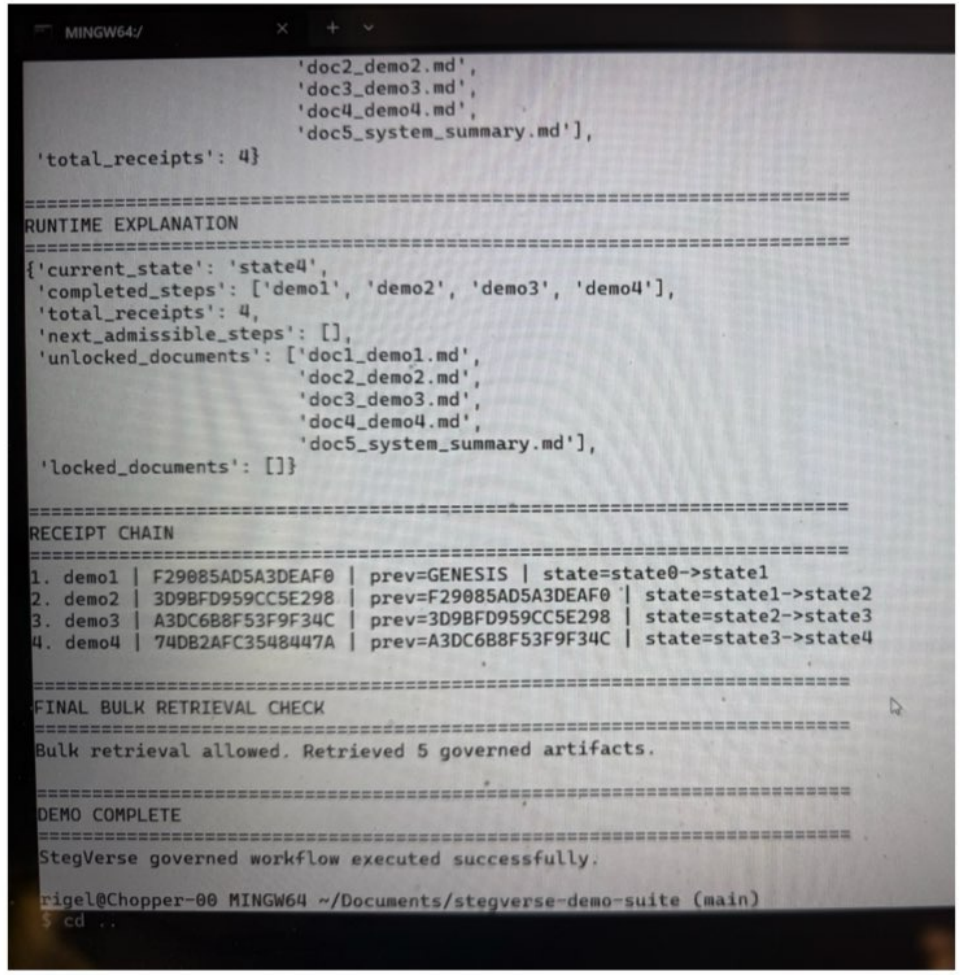
3) Governed Execution

# Appendix A — Execution Transcript

```
================================================================================
RUNNING DEMO4
================================================================================
State transition: state3 -> state4
Unlocked document: doc5_system_summary.md

================================================================================
FINAL STATUS
================================================================================
{'current_state': 'state4',
 'completed_steps': ['demo1', 'demo2', 'demo3', 'demo4'],
 'unlocked_documents': ['doc1_demo1.md',
                        'doc2_demo2.md',
                        'doc3_demo3.md',
                        'doc4_demo4.md',
                        'doc5_system_summary.md'],
 'total_receipts': 4}

================================================================================
RUNTIME EXPLANATION
================================================================================
{'current_state': 'state4',
 'completed_steps': ['demo1', 'demo2', 'demo3', 'demo4'],
 'total_receipts': 4,
 'next_admissible_steps': [],
 'unlocked_documents': ['doc1_demo1.md',
                        'doc2_demo2.md',
                        'doc3_demo3.md',
                        'doc4_demo4.md',
                        'doc5_system_summary.md'],
 'locked_documents': []}

================================================================================
RECEIPT CHAIN
================================================================================
1. demo1 | F29085AD5A3DEAF0 | prev=GENESIS | state=state0->state1
2. demo2 | 3D9BFD959CC5E298 | prev=F29085AD5A3DEAF0 | state=state1->state2
3. demo3 | A3DC6B8F53F9F34C | prev=3D9BFD959CC5E298 | state=state2->state3
4. demo4 | 74DB2AFC3548447A | prev=A3DC6B8F53F9F34C | state=state3->state4

================================================================================
```

4) Receipt-chain Generation

# Appendix A — Execution Transcript



```
MINGW64:/
                        'doc2_demo2.md',
                        'doc3_demo3.md',
                        'doc4_demo4.md',
                        'doc5_system_summary.md'],
 'total_receipts': 4}

==========================================================================
RUNTIME EXPLANATION
==========================================================================
{'current_state': 'state4',
 'completed_steps': ['demo1', 'demo2', 'demo3', 'demo4'],
 'total_receipts': 4,
 'next_admissible_steps': [],
 'unlocked_documents': ['doc1_demo1.md',
                        'doc2_demo2.md',
                        'doc3_demo3.md',
                        'doc4_demo4.md',
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3. demo3 | A3DC6B8F53F9F34C | prev=3D9BFD959CC5E298 | state=state2->state3
4. demo4 | 74DB2AFC3548447A | prev=A3DC6B8F53F9F34C | state=state3->state4

==========================================================================
FINAL BULK RETRIEVAL CHECK
==========================================================================
Bulk retrieval allowed. Retrieved 5 governed artifacts.

==========================================================================
DEMO COMPLETE
==========================================================================
StegVerse governed workflow executed successfully.

rigel@Chopper-00 MINGW64 ~/Documents/stegverse-demo-suite (main)
$ cd ..
```

5) Final Artifact Unlock Validation